

# Real-Time Transport in WebRTC

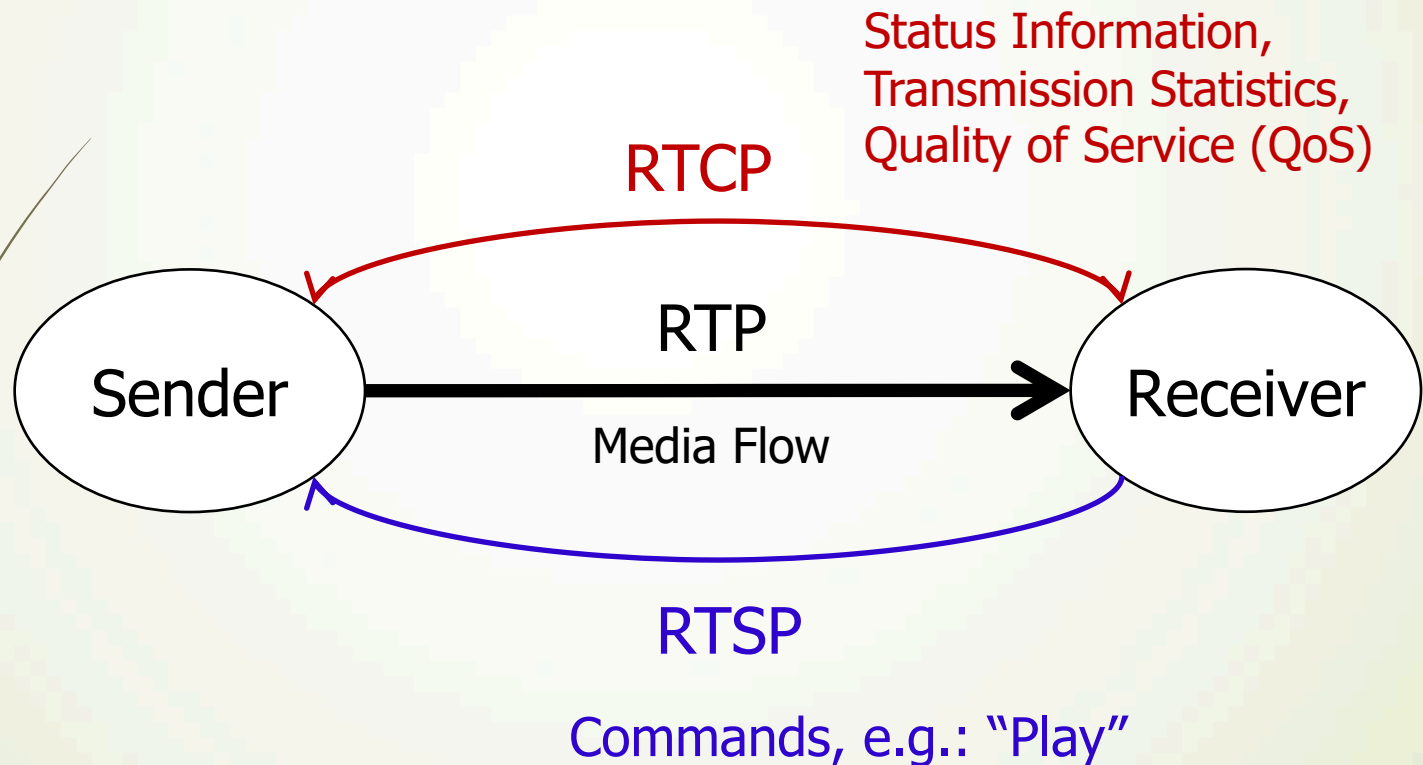
- RTP: **Real-Time Transport Protocol**
- RTCP: **RTP Control Protocol**
  - Published in 1996 as [RFC 1889](#), and superseded by [RFC 3550](#) in 2003
  - UDP, binary
  - Transmission direction:
    - RTP: from server to client (receiver)
    - RTCP: either way (SR, RR)

# Session Control in WebRTC

- **RTSP: Real-Time Streaming Protocol**
  - Published as [RFC 2326](#) in 1998
  - TCP, text
  - Transmission direction:
    - Initiation from client, response by server
  - “VCR”-type commands: PLAY, PAUSE, RECORD, TEARDOWN, ...
  - Session initiation: DESCRIBE, SETUP

# Protocol Suite

- Flow diagram: RTP, RTCP, RTSP



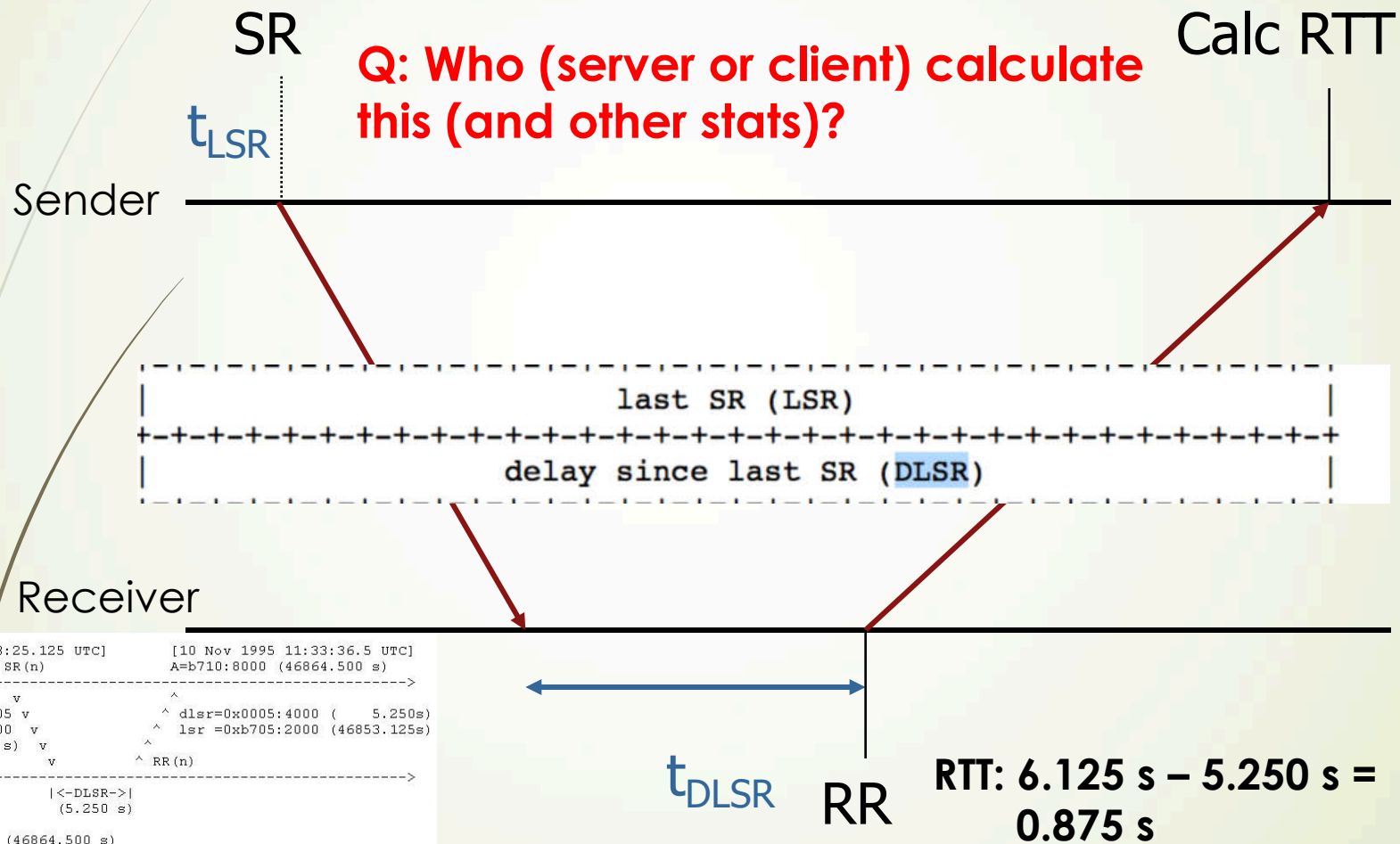
# Real-Time Control Protocol (RTCP) Overview

- Provides
  - receiver's feedback
  - network conditions
  - time synchronization
  - receiver's description
- Packet types
  - Sender's Report (SR)
  - Receiver's Report (RR)
  - Source Description (SDES)
  - Application Specific (APP)
  - BYE
- RCTP scaling: limit RTCP traffic to 5% of bandwidth; limit RR to 1.25%

# Receiver Reports Contain...

- Statistics
  - Number of lost packets
  - % of lost packets
  - Inter-arrival jitter
  - Timestamp of last SR
  - Delay since last SR
- Which can be used to derive
  - Packet Loss Rate
  - Interarrival Jitter
  - Round Trip Time

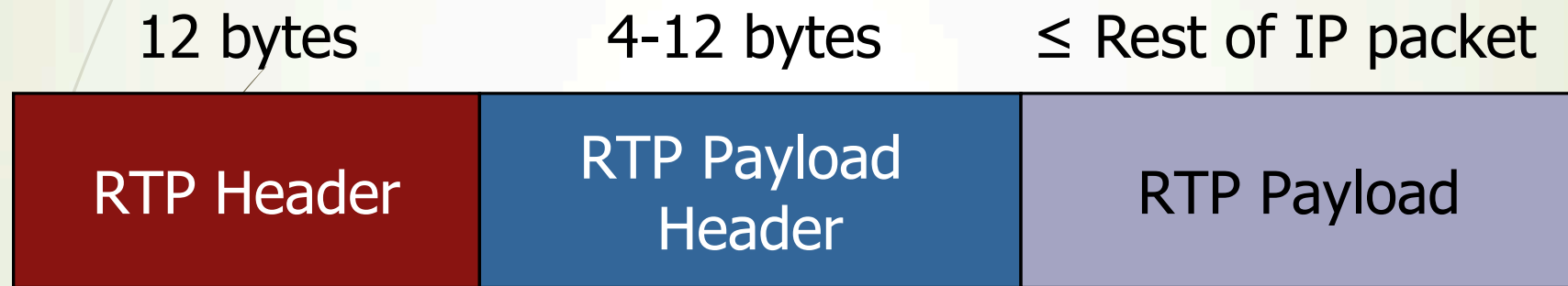
# Example: RTT Calculation



```
[10 Nov 1995 11:33:25.125 UTC]      [10 Nov 1995 11:33:36.5 UTC]
n          SR(n)                    A=b710:8000 (46864.500 s)
-----
v          ^
ntp_sec=0xb44db705 v          ^ dlsr=0x0005:4000 ( 5.250s)
ntp_frac=0x20000000 v          ^ lsr =0xb705:2000 (46853.125s)
(3024992005.125 s) v          ^
r          ^ RR(n)
-----
|<-DLSR->|
(5.250 s)

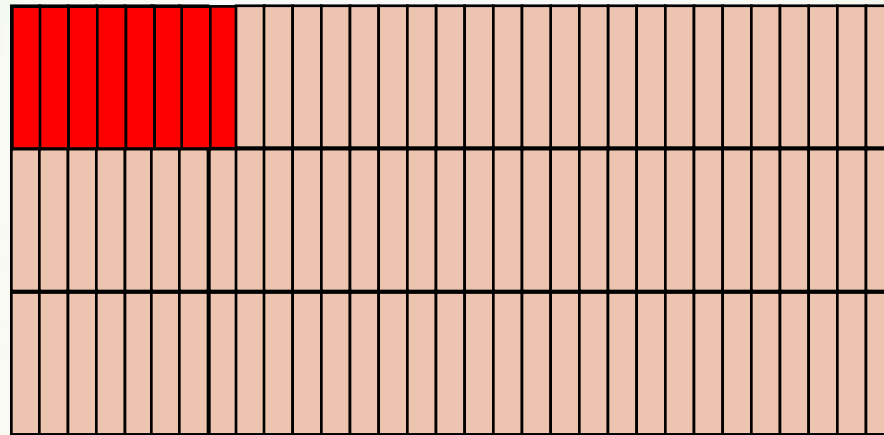
A      0xb710:8000 (46864.500 s)
DLSR -0x0005:4000 ( 5.250 s)
LSR  -0xb705:2000 (46853.125 s)
-----
delay 0x0006:2000 ( 6.125 s)
```

# Real-Time Transport (RTP) Protocol Overview



**Q: Why do we need RTP Payload Header?**

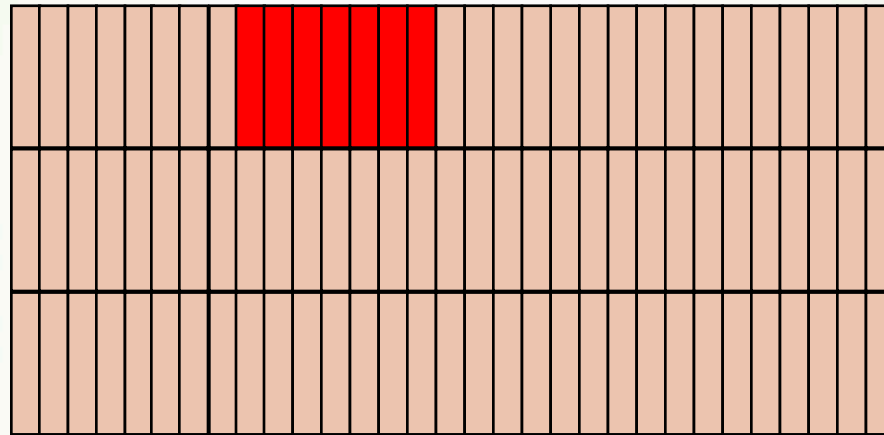
# RTP Header



- ➡ 9 bits: **protocol version, alignment, header extension, CSRC length, marker**

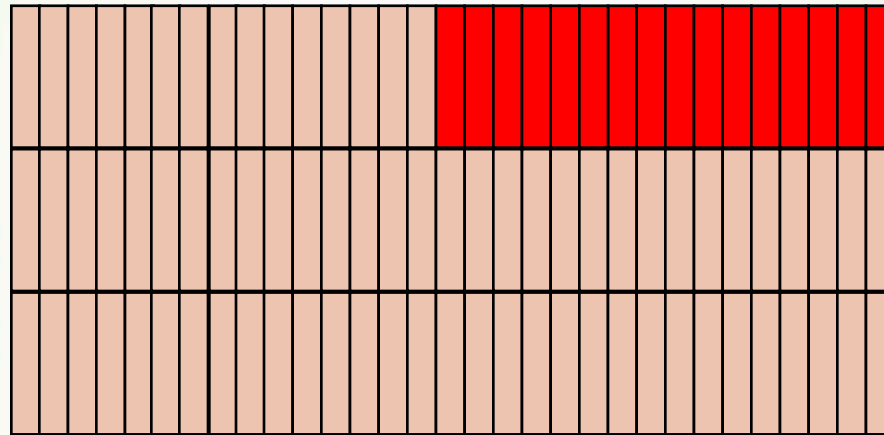


# RTP Header



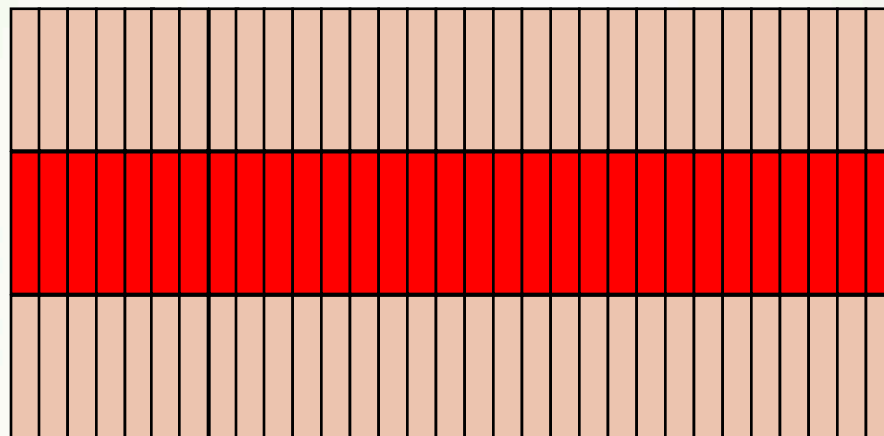
- **Payload type:** 7 bits
  - Identify content
  - E.g. 14: mp3; 32: MPEG-1

# RTP Header



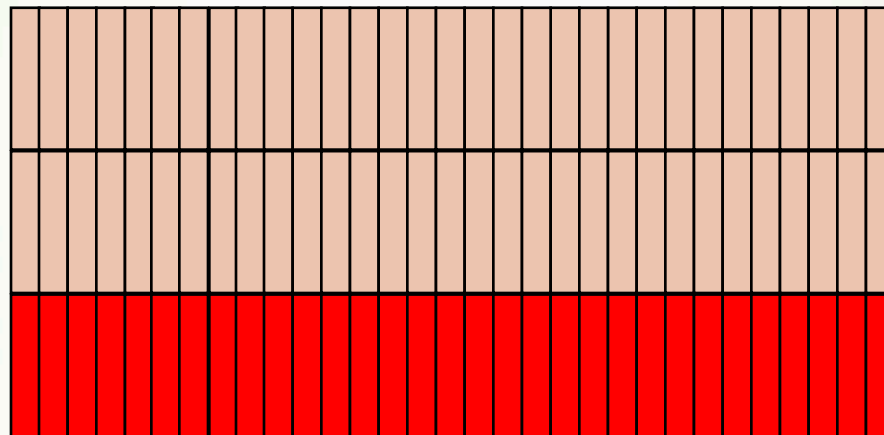
- **Sequence number:** 16 bits
  - Packet sequence number

# RTP Header



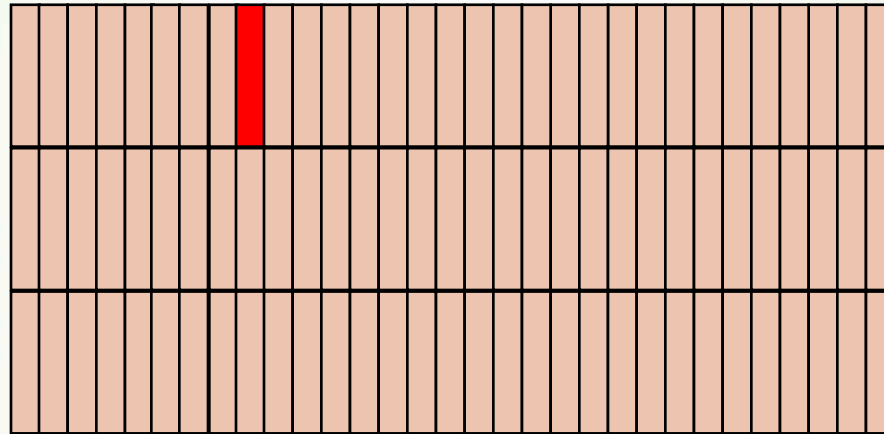
- **Media timestamp:** 32 bits
  - The instant when the first byte in this packet was captured
  - 90 kHz timestamp ( $90,000 = 1$  second)
- Note: timestamps for audio and video trackers are in different time scale ← Why?

# RTP Header



- **SSRC:** 32 bits
  - Random, unique in a session
  - Identifies a source (not host!)

# RTP Header



- **Marker bit:**
  - Depends on payload
  - E.g., beginning of frame

# What Clients Do After Receiving an RTP Packet?

- Check SSRC
  - New source?
  - Existing source? which one?
- Check payload type
  - Has format been changed?
  - Which decoder should I use?

# Real-Time Streaming Protocol (RTSP) Overview

- **Application-level protocol** for establishing and controlling media sessions with real-time properties between end points control
- Simple, text-based
- Published in RFC 2326 (1998)
- Uses TCP
- Standard port: 554
- Allows VCR-type commands:
  - DESCRIBE, SETUP, PLAY, TEARDOWN, PAUSE, RECORD, OPTIONS

# Sample RTSP Request/Response

DESCRIBE rtsp://genesis/hackers.mov RTSP/1.0

RTSP/1.0 200 OK

Server: QTSS/v96

Cseq:

Content-Type: application/sdp

Content-Base: rtsp://genesis/hackers.mov/

Content-length: 179

v=0

s=hackers.mov

u=http://genesis.usc.edu/

e=admin@genesis.usc.edu

c=IN IP4 128.125.163.19

a=control:/

a=range:npt=0-3714.90167

m=audio 0 RTP/AVP 10

a=control:trackID=2



# Application Level Framing

- How/When to send/recv?
  - Let the application decide, not the protocol stack
- Applications know
  - How to reorder packets
  - Whether to ignore loss
  - Which packet to retransmit
- Application Data Unit (ADU)
  - Can be processed individually, even out-of-order
  - Unit of error-recovery
  - 8-Bit PCM audio: 1 ADU = 1 Byte
  - MPEG(1, 2, ...) Video: 1 ADU = 1 Slice
- Ideally, 1 ADU in 1 packet ← **Why?**

# Agenda

- DASH Overview
- ABR: Adaptive Bitrate Allocation
- WebRTC and Underlying Protocols (RTP, RTCP, RTSP)
- DASH versus RTP

# DASH versus RTP

|                    | <b>DASH</b>      | <b>RTP</b>               |
|--------------------|------------------|--------------------------|
| Transport Protocol | TCP              | UDP                      |
| Latency            | long             | short                    |
| Media unit         | segment          | packet                   |
| Topology           | client-server    | peer-to-peer             |
| Codec              | many             | many                     |
| Caching            | yes (web caches) | no                       |
| Applications       | VoD, 1-way live  | VoIP, video conferencing |

# Questions

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